

EVIDENCE APPENDIX

Is the problem real?

Evidence and validation so far

CURRENT VERDICT

**Credible problem.
Unproven Bataa fit.**

EVIDENCE LADDER

-
- 01 Direct**
Bataa conversations + prototype
 - 02 External**
Research and market behavior
 - 03 Unproven**
Retention, outcomes, payment

The honest answer has two parts

External evidence supports the problem and mechanism. Bataa-specific proof is still ahead.

WE CAN DEFEND

The problem mechanism

1

BEHAVIOR

Learners already use on-demand online resources.

2

PEDAGOGY

Practice and informative feedback improve learning.

WE CANNOT CLAIM

Bataa product-market fit

1

UNKNOWN

How many learners feel this pain often enough.

2

UNKNOWN

Whether Bataa improves outcomes or earns payment.

Decision-quality statement: defend the thesis, then run a measured pilot.

Bataa has a directional signal from real learners

The team heard a recurring support gap at Horus University and built around it.

FOUNDER-REPORTED

**“The question arrives
after the lesson.”**

Horus University conversations surfaced late-night questions, hesitation about asking “basic” questions, and concern about cost or crowded courses.

What exists today

- 1 DISCOVERY**
A problem statement grounded in conversations, not a quantified survey.
- 2 PROTOTYPE**
A mobile daily-task loop and a desktop guidance concept already exist.
- 3 MISSING RECORD**
No verified participant count, dates, recruitment method, or transcripts were supplied.

Use this as discovery evidence. Do not present it as prevalence data.

Egypt survey: students wanted more interaction

A large medical-student survey gives local context for the support gap, with important limits.

4,935

responses

26

Egyptian medical schools

>50%

wanted more interactive sessions

63%

rated recorded video tutorials most effective

54.6%



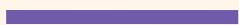
said online education was not as effective as face-to-face

23%



reported bad internet connection

17%



found it difficult to commit because of home distractions

18%



reported less communication than face-to-face learning

Limit: pandemic-era medical students, social-media recruitment, descriptive self-report. This is local context, not Bataa demand.

Self-directed coding learning is already normal

The opportunity is to make the next action more guided, not to invent online learning from zero.

65,437

survey responses

185

countries represented

82%

say online resources are their top choice

37%

say AI helps them learn to code

What the survey shows

83.9%

technical documentation

80.3%

Stack Overflow

37.0%

AI tools

29.3%

interactive tutorials

WHAT THIS SUPPORTS

On-demand learning behavior already exists.

It does not prove novice demand in Egypt, Bataa retention, or a willingness to pay.

Practice beats passive exposure

The research base supports a small action, a check, and another attempt.

225

undergraduate STEM studies
in a meta-analysis

+0.47 SD

exam performance with active learning

FAILURE RISK

1.95×

odds of failure under traditional lecture
vs active learning

IMPORTANT DESIGN DETAIL

149-student physics study

Active instruction improved actual learning (+0.46 SD), while students felt they learned less (−0.56 SD).

Feedback works when it carries information

Correctness alone is not enough. The learner needs a reason and a next move.

435

studies in a feedback
meta-analysis

$d = 0.48$

medium overall effect on learning, with
substantial variation by feedback content

BATAA DESIGN RULE

01

Explain what happened

Name the concept or step behind the result.

02

Show the next move

Offer a hint or smaller action, not a wall of text.

03

Protect the retry

Let a learner try privately without public embarrassment.

Caveat: 17% of feedback effects were negative in the review. Format and information content matter.

Contextual AI has an early evidence signal

Adjacent product experiments suggest that relevant history can improve the next step.

GITHUB COPILOT

95 professional developers

78% vs 70%

completed the same JavaScript HTTP-server task

55% faster

1h11 vs 2h41 average completion time

ADJACENT EVIDENCE

KHANMIGO

>15M tutoring threads

+3.4%

recent problem history improved next-item correctness

+2.7%

unmastered prerequisite review improved next-item correctness

+6.1%

combined gain in next-item correctness

+5.09%

prior-24h context increased cognitive engagement

Both are adjacent evidence, not novice coding proof.

What the evidence supports, and what it leaves open

A credible thesis is not the same thing as product-market fit.

QUESTION	STATUS	EVIDENCE SAYS	STILL UNPROVEN
Is there a real need for on-demand support?	SUPPORTED	Online resources dominate coding learning; Egyptian students wanted more interaction.	How often Bataa's target learners feel it.
Does practice and feedback improve learning?	SUPPORTED	225-study active-learning meta-analysis; 435-study feedback meta-analysis.	Which Bataa feedback designs transfer to coding.
Can contextual AI reduce friction?	PROMISING	Copilot and Khanmigo experiments show adjacent performance gains.	Novice coding, trust, safety, and long-term learning.
Do Bataa learners return and finish?	UNKNOWN	No Bataa behavioral cohort yet.	Activation, D7 retention, completion, transfer.
Will users or partners pay?	UNKNOWN	No willingness-to-pay or paid pilot evidence yet.	Price, conversion, institution/creator demand.

The next proof should be behavioral

Turn the thesis into a small, measured pilot before scaling the story.

PILOT SHAPE

01

15–20 discovery interviews

Record consented sessions, recruit across beginner skill levels, code recurring moments.

02

20–30 learner pilot

Run one beginner path for 14 days with simple daily tasks.

03

Comparison condition

Compare contextual feedback against the current lesson flow or a baseline.

MEASURE BEFORE SCALE



Activation

first task started



Task success

time-to-first-success



Feedback loop

retry after a hint



Retention

D1 and D7 return



Transfer

delayed task without help



Commercial

willingness to pay / partner ask

Predefine pass / fail thresholds before the pilot starts.

We can defend the problem and the product thesis.

We cannot yet claim product–market fit.

REAL PROBLEM?

Credible

External research and local context converge.

BATAA FIT?

Unproven

No measured cohort or payment signal yet.

NEXT DECISION

Run the pilot

Measure behavior before scaling claims.

Sources used in this appendix

Primary links are preserved in speaker notes for each slide.

Peer-reviewed / public research

- 1 Freeman et al. (2014), PNAS · active learning meta-analysis
- 2 Deslauriers et al. (2019), PNAS · actual vs perceived learning
- 3 Wisniewski et al. (2020), Frontiers · feedback meta-analysis
- 4 Mortagy et al. (2022), BMC Medical Education · Egypt survey
- 5 Gillett-Swan (2017), Journal of Learning Design · online isolation

Official / company research

- 1 Stack Overflow Developer Survey (2024)
- 2 GitHub Copilot controlled experiment
- 3 Khan Academy Khanmigo product tests (May 2026)
- 4 Mimo, freeCodeCamp, and Codecademy were reviewed as market signals, not used as core proof
- 5 Internal Bataa discovery account · founder-reported

Limitations: selection bias · self-report · generalizability · vendor affiliation · no Bataa behavioral cohort